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CH.SC.U4CSE24222

Week – 7

Design and Analysis of Algorithm(23CSE211)

Job Sequencing

Write a c program for Job Sequencing

Code:

```
#include <stdio.h>
#include <stdlib.h>
typedef struct {
    int id;
    int deadline;
    int profit;
} Job;
void jobScheduling(Job jobs[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (jobs[j].profit < jobs[j + 1].profit) {
                Job temp = jobs[j];
                jobs[j] = jobs[j + 1];
                jobs[j + 1] = temp;
            }
        }
    }
    int maxDeadline = 0;
    for (int i = 0; i < n; i++) {
        if (jobs[i].deadline > maxDeadline) {
            maxDeadline = jobs[i].deadline;
        }
    }
    int *result = (int *)malloc(maxDeadline * sizeof(int));
    int *slot = (int *)malloc(maxDeadline * sizeof(int));
    int totalProfit = 0;
    for (int i = 0; i < maxDeadline; i++) {
        slot[i] = 0;
```

```

}
for (int i = 0; i < n; i++) {
    for (int j = jobs[i].deadline - 1; j >= 0; j--) {
        if (slot[j] == 0) {
            result[j] = i;
            slot[j] = 1;
            totalProfit += jobs[i].profit;
            break;
        }
    }
}

printf("\nJob Sequence: ");
for (int i = 0; i < maxDeadline; i++) {
    if (slot[i]) {
        printf("%d ", jobs[result[i]].id);
    }
}

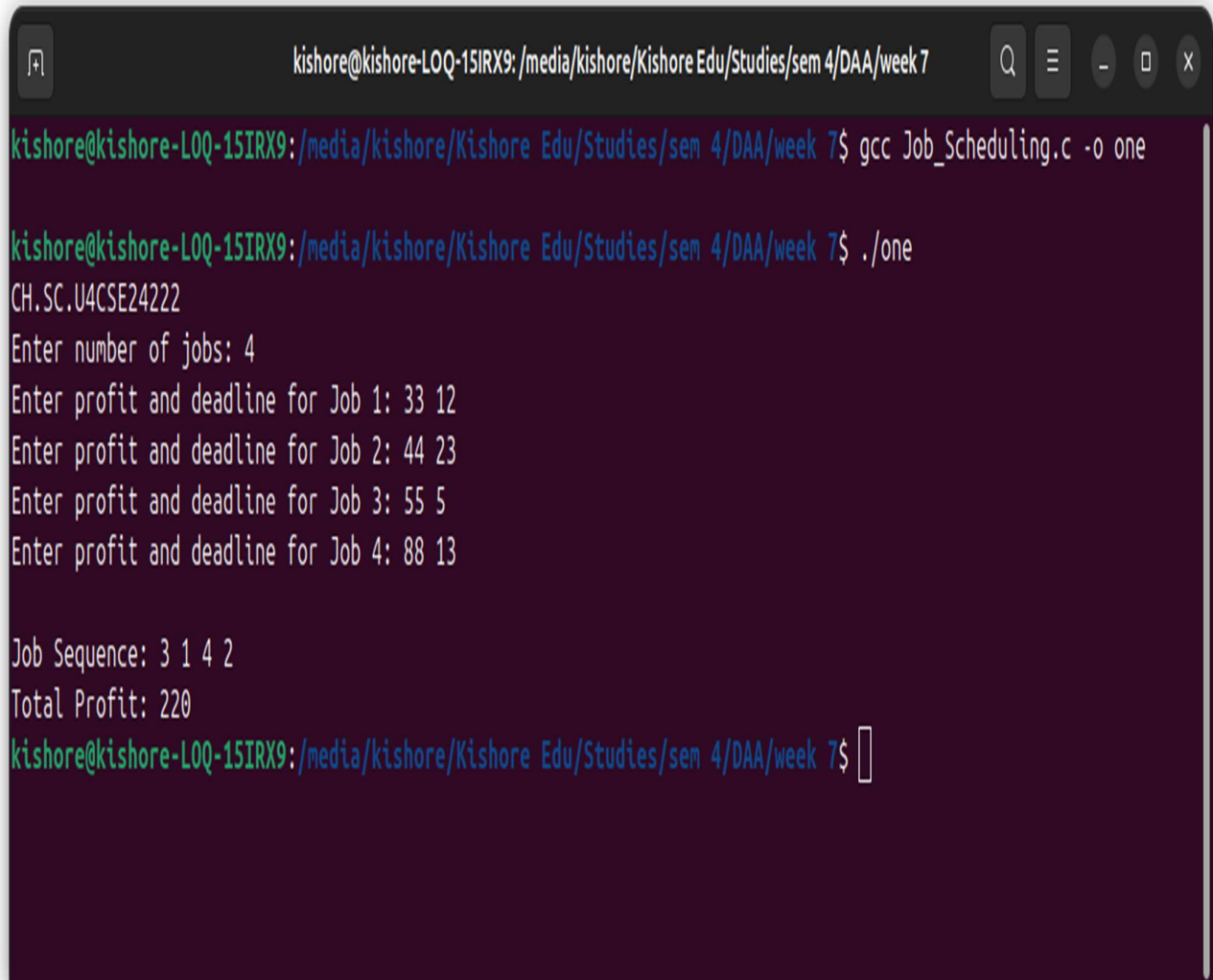
printf("\nTotal Profit: %d\n", totalProfit);
free(result);
free(slot);
}

int main() {
    printf("CH.SC.U4CSE24222\n");
    int n;
    printf("Enter number of jobs: ");
    scanf("%d", &n);
    Job jobs[n];
    for (int i = 0; i < n; i++) {
        jobs[i].id = i + 1;
        printf("Enter profit and deadline for Job %d: ", jobs[i].id);
    }
}

```

```
    scanf("%d %d", &jobs[i].profit, &jobs[i].deadline);  
}  
jobScheduling(jobs, n);  
return 0;  
}
```

OUTPUT:



```
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 7  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 7$ gcc Job_Scheduling.c -o one  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 7$ ./one  
CH.SC.U4CSE24222  
Enter number of jobs: 4  
Enter profit and deadline for Job 1: 33 12  
Enter profit and deadline for Job 2: 44 23  
Enter profit and deadline for Job 3: 55 5  
Enter profit and deadline for Job 4: 88 13  
  
Job Sequence: 3 1 4 2  
Total Profit: 220  
kishore@kishore-LOQ-15IRX9: /media/kishore/Kishore Edu/Studies/sem 4/DAA/week 7$
```